forum acusticum 2026
8-12 September · Graz, Austria

FROM TRANSMISSION LINE MODELS TO STUART–LANDAU OSCILLATORS: A HOPF-BASED DESCRIPTION OF COCHLEAR DYNAMICS

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This paper is a revised version of the authors' submitted manuscript that was peer-reviewed by at least two anonymous reviewers.

Abstract

Over the past decades, numerous models of the cochlear partition have been proposed to explain key features of cochlear mechanics, including traveling waves, nonlinear amplification, and compressive responses. Among them, the transmission-line model introduced by Geoffrey Zweig and extended by others represents the cochlea as a distributed system of coupled resonant elements interacting through the cochlear fluid and successfully reproduces the spatiotemporal dynamics of the basilar membrane.

Another widely used framework describes active auditory processes as nonlinear oscillators operating near a Hopf bifurcation. The Stuart–Landau oscillator (SLO), the normal form of a Hopf bifurcation, captures essential properties such as self-sustained oscillations, nonlinear amplification, and compressive responses, and has therefore been widely used to model active hair-cell dynamics and cochlear amplification.

In this study, we analytically demonstrate that the local dynamics of the transmission-line model undergo a Hopf bifurcation and can be reduced to the Hopf normal form of an SLO. We also present a numerical simulation framework for the displacement of the basilar membrane, highlighting its similarity to a network of coupled oscillators and to a transmission-line model. These results suggest that the dynamics of the cochlear partition can be interpreted as an ensemble of coupled Hopf oscillators.

Keywords: *MANDATORY, 3 to 5 keywords, separated by a comma, italic, not bold, no capitalized letters*

1. Introduction

The mammalian cochlea performs a remarkable mechanical signal-processing task, transforming sound-induced pressure waves into frequency-selective neural responses. Incoming acoustic stimuli generate pressure differences across the cochlear partition that propagate as traveling waves along the basilar membrane. The resulting vibrations are shaped by the complex micromechanics of the organ of Corti, where the basilar membrane, tectorial membrane, and sensory hair cells interact to produce the characteristic frequency tuning and nonlinear response of the auditory system. These dynamics are not purely passive; instead, they involve an active amplification mechanism, primarily mediated by outer hair cells, that enhances sensitivity to weak stimuli and produces the compressive nonlinearities observed in cochlear responses [1], [2].

To describe the propagation of mechanical waves in the cochlea, several mathematical and computational models have been proposed. One influential class of models represents the cochlea as a transmission line, in which the basilar membrane and organ of Corti are modeled as spatially distributed mechanical impedances coupled through the cochlear fluid. Early work by Zweig [3] and later by Shera [4] formulated the cochlea as a distributed system of coupled resonant elements whose interaction with the cochlear fluid produces the traveling-wave behavior observed experimentally. In this framework, the inertia of the cochlear fluid provides longitudinal coupling between adjacent elements. At the same time, the local mechanical impedance determines the frequency-dependent response of each section of the cochlear partition. Transmission-line models have proven highly successful in reproducing key features of cochlear mechanics, including the spatial progression of the traveling wave and the frequency-dependent tuning along the basilar membrane [5].

More recent work has extended this framework to incorporate the cochlea's active nonlinear mechanics. In particular, models developed by Verhulst and collaborators [6], [7] introduced an important modification to the classical transmission-line formulation by allowing the poles of the local mechanical transfer function to shift with stimulus level. Earlier transmission-line models were largely linear systems with fixed pole locations determining the resonance characteristics of each cochlear partition element. In contrast, the model from [4] introduces level-dependent pole shifting, effectively modeling the cochlear amplifier and generating compressive non-

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linear responses consistent with experimental observations. In this formulation, each longitudinal section of the cochlea is represented by a local filter whose pole locations determine the basilar-membrane response's resonance frequency and damping. Allowing these poles to move in the complex plane as a function of stimulus level provides a computationally efficient mechanism for incorporating active cochlear amplification into the transmission-line framework while preserving the traveling-wave propagation along the basilar membrane.

An alternative theoretical perspective frames cochlear amplification as a nonlinear dynamical system operating near a Hopf bifurcation. In this framework, the active elements of the cochlea are assumed to operate near a dynamical instability, where small perturbations can be strongly amplified. Near such a bifurcation, the dynamics can be described by the SLO, which represents the normal form of the Hopf bifurcation. This model has been derived from a damped spring–mass system by Vilfan and Duke [8] and has been successfully compared to spontaneous otoacoustic emissions (SOAEs) observed in lizards. More generally, Hopf-type oscillator models have been widely used to describe the active mechanics of hair bundles and outer hair cells, and to explain key phenomena such as sharp frequency tuning and SOAEs [9], [10], [11].

Despite the success of both modeling approaches, the relationship between transmission-line descriptions of cochlear wave propagation and oscillator-based models of cochlear amplification remains largely conceptual. However, Hulst et al. [12] have shown that it "has the same form as the equation of motion of an active oscillator operating near a Hopf bifurcation in its linear regime of responsiveness". In particular, the level-dependent pole shifting introduced in modern transmission-line models suggests that the system may approach a dynamical instability in which the effective damping of the cochlear partition is reduced. This observation motivates a dynamical analysis of the transmission-line model within the framework of nonlinear systems theory.

Motivated by these considerations, the present study investigates the dynamical properties of the transmission-line model. We demonstrate that the local dynamics undergo a Hopf bifurcation and can therefore be reduced to the Stuart–Landau normal form. This establishes a direct connection between transmission-line models and Hopf-type oscillator descriptions of cochlear amplification. To highlight this correspondence, a numerical simulation framework is presented to compare basilar membrane displacement between an ensemble of coupled Stuart–Landau oscillators and transmission-line model results.

2. Derivation of the Hopf Normal Form

One local element of the transmission line model from [7] has the form:

$$\begin{aligned} v &= \dot{x} \\ \dot{v} &= -\omega^2 x - \hat{\mu} v - \frac{1}{m}\Psi(x(t-\tau))x(t-\tau). \end{aligned} \tag{1}$$

Define

$$\begin{aligned} \omega &= \sqrt{\frac{k}{m}}, \\ \hat{\mu} &= \frac{d}{m}, \\ x_\tau &= x(t-\tau). \end{aligned} \tag{2}$$

Then

$$\dot{v} = -\omega^2 x - \hat{\mu} v - \frac{1}{m}\Psi(x_\tau)x_\tau. \tag{3}$$

The equilibrium is

$$\begin{aligned} v &= 0, \\ \dot{v} &= 0. \end{aligned} \tag{4}$$

The nonlinear function Ψ is expanded as

$$\begin{aligned} \dot{v} &= -\omega^2 x - \hat{\mu} v \\ &-\frac{1}{m}\left(\Psi(0)x_\tau + \Psi'(0)x_\tau^2 + \Psi''(0)\frac{x_\tau^3}{2}\right). \end{aligned} \tag{5}$$

2.1. Hopf Conditions

For delay differential equations, solutions of the linearized system are of the form $Ae^{\lambda t}$ with characteristic roots (roots)

$$\lambda = \sigma + j\kappa. \tag{6}$$

A Hopf bifurcation occurs if two conditions are satisfied. First, a pair of complex conjugate roots crosses the imaginary axis. Second, this crossing is transversal with respect to a system parameter, as noted in [13].

The first condition requires that the roots with maximal real part are purely imaginary:

$$H_1 : \max \quad \Re(\lambda) = \lambda_0 = \pm j\kappa. \tag{7}$$

The second condition ensures that the real part of the roots changes with the parameter:

$$H_2 : \Re\left(\frac{\partial \lambda}{\partial \hat{\mu}}(\hat{\mu}_0)\right) \neq 0. \tag{8}$$

For the analysis, we use the exponential ansatz.

$$\begin{aligned} x(t) &= Ae^{\lambda t} \\ v(t) &= A\lambda e^{\lambda t} \\ \dot{v}(t) &= A\lambda^2 e^{\lambda t}. \end{aligned} \tag{9}$$

2.1.1. Condition H_1

Linearizing around the equilibrium and substituting the exponential ansatz gives the characteristic equation.

$$h(\lambda) = -\lambda^2 - \omega^2 - \hat{\mu}\lambda - \frac{\Psi(0)}{m}e^{-\lambda\tau}. \tag{10}$$

At the Hopf bifurcation, the separation of real and imaginary parts gives

$$\begin{aligned} -\frac{\Psi(0)}{m}\cos(\omega\tau) &= 0 \\ -\hat{\mu}_0\omega + \frac{\Psi(0)}{m}\sin(\omega\tau) &= 0. \end{aligned} \tag{11}$$

Hence

$$\cos(\omega\tau) = 0, \tag{12}$$

which implies $\omega\tau = \frac{\pi}{2} + k\pi, \quad k \in \mathbb{Z}$. The corresponding parameter value is

$$\hat{\mu}_0 = \pm\frac{\Psi(0)}{m\omega}\sin(\omega\tau). \tag{13}$$

Choosing the branch

$$\omega\tau_0 = \frac{\pi}{2} \tag{14}$$

yields

$$\hat{\mu}_0 = \frac{\Psi(0)}{m\omega}. \tag{15}$$

At this point, the roots are $\lambda = +-j\omega$. The characteristic roots depend continuously on τ. At $\tau = 0$, all roots satisfy $\Re(\lambda) < 0$ since

$$\lambda = \frac{-\hat{\mu}_0 \pm \sqrt{\hat{\mu}_0^2 - 4\left(\omega^2 + \frac{\Psi(0)}{m}\right)}}{2}, \tag{16}$$

and therefore

$$\sigma_{\tau=0} = -\frac{\hat{\mu}_0}{2} < 0. \tag{17}$$

By continuity of the roots with respect to τ, and since only one simple pair reaches the imaginary axis, all remaining roots satisfy $\Re(\lambda) < 0$ at τ_0. Hence, H1 holds.

2.1.2. Condition H_2

The roots satisfy

$$h(\lambda(\hat{\mu}), \hat{\mu}, \tau_0) = 0. \tag{18}$$

Differentiating with respect to $\hat{\mu}$ yields

$$\frac{\partial h}{\partial \hat{\mu}} = \frac{\partial h}{\partial \lambda}\frac{\partial \lambda}{\partial \hat{\mu}} + \frac{\partial h}{\partial \hat{\mu}} = 0, \tag{19}$$

so that

$$\frac{\partial \lambda}{\partial \hat{\mu}} = -\frac{\lambda}{-2\lambda - \hat{\mu} + \tau_0\frac{\Psi(0)}{m}e^{-\lambda\tau_0}}. \tag{20}$$

Evaluating at $\lambda = j\omega$ gives

$$\frac{\partial \lambda}{\partial \hat{\mu}}(\hat{\mu}_0) = \frac{-j\omega}{-\hat{\mu}_0 - j\left(2\omega + \tau_0\frac{\Psi(0)}{m}\right)}. \tag{21}$$

If $\omega \neq 0$ and $2\omega + \tau_0\frac{\Psi(0)}{m}) \neq 0$, it follows that

$$\Re\left(\frac{d\lambda}{d\hat{\mu}}(\hat{\mu}_0)\right) \neq 0. \tag{22}$$

Thus, condition H_2 is satisfied.

This shows that the system has a Hopf bifurcation at $\tau = \frac{\pi}{2\omega}$ and $\hat{\mu} = \frac{\Psi(0)}{m\omega}$.

2.2. Center Manifold Reduction

Near the bifurcation, only the critical eigenmodes contribute to the dynamics, as all other modes decay exponentially and are therefore slaved to the critical modes (as shown in Section 2.1.1). Consequently, the system can be reduced to a two-dimensional center eigenspace. The solution can therefore be written as

$$\begin{aligned} x &= A(e^{j\omega t} + e^{-j\omega t}) \\ v &= j\omega A(e^{j\omega t} - e^{-j\omega t}) \end{aligned} \tag{23}$$

which can be written as a complex variable

$$z = x - \frac{j}{\omega}v. \tag{24}$$

Substituting into the nonlinear equation yields

$$\begin{aligned} \dot{z} - \dot{\bar{z}} = &\left(j\omega - \hat{\mu} + j\frac{\Psi(0)}{m\omega}e^{j\omega\tau}\right)z \\ &+\left(j\omega + \hat{\mu} + j\frac{\Psi(0)}{m\omega}e^{j\omega\tau}\right)\bar{z} \\ &+\frac{\Psi'(0)}{m\omega}(z_\tau^2 + 2z_\tau\bar{z}_\tau + \bar{z}_\tau^2) \\ &+\frac{\Psi''(0)}{2\omega m}(z_\tau^3 + 3z_\tau^2\bar{z}_\tau + 3z_\tau\bar{z}_\tau^2 + \bar{z}_\tau^3). \end{aligned} \tag{25}$$

Only resonant terms that match the critical eigenfrequency contribute to the reduced dynamics, while non-resonant terms are absorbed into higher-order correction terms. Projecting onto the center eigenspace yields

$$\begin{aligned} \dot{z} = &\left(-\hat{\mu} + i\omega + j\frac{\Psi(0)}{m\omega}e^{-i\omega\tau}\right)z \\ &+\frac{3}{2\omega m}\Psi''(0)e^{-j\omega\tau}|z|^2 z. \end{aligned} \tag{26}$$

Introduce the coefficients

$$\begin{aligned} \hat{\mu}_\Psi + j\omega_\Psi &= j\frac{\Psi(0)}{m\omega}e^{j\omega\tau} \\ \alpha + j\beta &= \frac{3}{2\omega m}O''(0)e^{j\omega\tau}. \end{aligned} \tag{27}$$

The Hopf normal form becomes

$$\dot{z} = (\mu + j\omega_0)z + (\alpha + j\beta)|z|^2 z. \tag{28}$$

with the parameter

$$\begin{aligned} \mu &= \hat{\mu}_\Psi - \hat{\mu} \\ \omega_0 &= \omega + \omega_\Psi \end{aligned} \tag{29}$$

Evaluating at the bifurcation point yields

$$\dot{z} = (j\omega)z + \left(-j\frac{3}{2\omega m}\Psi''(0)\right)|z|^2 z, \tag{30}$$

which is consistent with the local normal form at the Hopf bifurcation. This shows that the transmission line model reduces to an SLO in a neighborhood of the bifurcation.

2.3. Numerical Simulation of Stuart–Landau Oscillators

The oscillator dynamics derived in the previous section are simulated numerically using ensembles of SLO. To allow both deterministic and stochastic simulations within the same framework, the system is formulated as a stochastic differential equation (SDE). The general form used in the simulations is

$$\begin{aligned} \dot{z}_i = &(\mu_i + i\omega_{0,i})z_i - (1 + i\beta_i)|z_i|^2 z_i \\ &+C_i(z) + \nu_i(t) + F(t). \end{aligned} \tag{31}$$

Here $z_{i(t)}$ denotes the complex oscillator state and μ, ω_i, and β are the parameters introduced previously. The term α is always set to 1 because its influence can be modeled by μ [14]. The term $\nu_{i(t)}$ represents a stochastic contribution modeled as a Wiener process, while $C_{i(z)}$ and $F(t)$ denote coupling and external forcing terms. Formulating the system as a stochastic differential equation allows stochastic perturbations to be included in the oscillator dynamics when required. The stochastic term $\nu_{i(t)}$ can be used to introduce white-noise fluctuations

acting on the oscillator state. When this term is disabled, the system reduces to a deterministic ordinary differential equation. In this way, deterministic and stochastic simulations can be treated within a unified numerical framework.

Coupling between oscillators is introduced through the nearest-neighbor coupling used by Epp et al. [15] with the term $C_{i(z)}$, which depends on the states of neighboring oscillators. In the implemented model, oscillators are arranged in a spatial array, and each oscillator interacts with its two nearest neighbors. The coupling term is given by

$$\begin{aligned} C_i(z) = k_{i-1}\Re(z_{i-1}) + k_{i+1}\Re(z_{i+1}) \\ +j(d_{i-1}\Im(z_{i-1}) + d_{i+1}\Im(z_{i+1})). \end{aligned} \tag{32}$$

This interaction introduces dynamical dependencies between oscillators in the ensemble. In practice, the combined influence of neighboring oscillators can lead to irregular perturbations of individual oscillator trajectories. Such effects may resemble stochastic fluctuations, which motivates the use of a stochastic differential equation formulation even in simulations where explicit noise terms are small or absent.

The simulations are implemented in Julia using the DifferentialEquations.jl framework. The oscillator system is integrated over the specified time interval using solvers designed for stochastic differential equations. In the present implementation, the Euler–Maruyama method is employed. This method advances the deterministic part of the dynamics while adding Wiener increments that represent stochastic perturbations at each integration step.

External forcing is implemented through the time-dependent input signal $F_{i(t)}$. The forcing signal is typically provided as a discrete time series. Because the numerical solver uses adaptive step sizes, the system may be evaluated at time points that do not coincide with the input signal's sampling times. To ensure the forcing term can be evaluated consistently during integration, the discrete signal is converted to a continuous function via linear interpolation. This allows the solver to evaluate the forcing term at arbitrary intermediate time points during the simulation.

Before performing the main simulation, a short preliminary integration is performed to generate suitable initial conditions for the oscillator states. Starting the simulation from arbitrary initial values can introduce unwanted dynamics, such as long convergence times, that are unrelated to the system's steady-state behavior. Therefore, the oscillator system is first simulated for a short time interval without external forcing. The final state of this preliminary simulation is then used as the initial condition for the actual simulation. This procedure ensures that the oscillator network begins the main simulation from a dynamically consistent state.

3. Results

To demonstrate the model's capability to reproduce cochlear partition responses, a simulation was performed using an ensemble of 600 SLOs, logarithmically spaced between 200Hz and 16kHz, according to the frequency-place map proposed by Greenwood [16]. The oscillators were coupled via nearest-neighbor interactions, and no stochastic noise was included. To account for the amplification of the outer and middle ear, the input signal was amplified by 20dB. The simulation allows observation of key cochlear features, including frequency selectivity, nonlinear compression, and traveling wave propagation.

It should be noted that the last 50 oscillators (corresponding to frequencies between 8192Hz and 16kHz) did not reach a steady-state response within the simulation time. This behavior was observed independently of the specific choice of characteristic frequencies and is likely due to coupling-induced boundary effects, leading to altered dynamics at the ends of the oscillator array compared to its central region. Therefore, the responses of these oscillators were excluded from the following analysis. Nevertheless, this phenomenon is of potential interest and warrants further investigation, although it is beyond the scope of this work.

In Figure 1, the steady-state response of the oscillator ensemble to sinusoidal stimuli at 1kHz with varying SPL is shown. The results exhibit pronounced frequency selectivity at low input levels (e.g., 10dB and 50dB SPL), which decreases with increasing stimulus amplitude. In addition, the peak response shifts toward lower characteristic frequencies as the input level increases. Furthermore, a clear compressive nonlinearity is observed for input levels above approximately 50dB SPL.

In Figure 2, the displacement of a subset of the oscillator ensemble during the transient phase prior to steady-state is shown. The results illustrate the propagation of a traveling wave along the oscillator array, with the response peak moving from basal (high-frequency) to apical (low-frequency) regions over time. The amplitude evolves along the array, reaching its maximum near the oscillator whose characteristic frequency is closest to the stimulus frequency.

4. Discussion

From the derivation, it follows that the transmission line model can be reduced to an SLO near the Hopf bifurcation. This reduction is valid because the delayed feedback term effectively introduces a velocity-dependent component into the dynamics. In particular, for a delay satisfying $\omega\tau \approx \frac{\pi}{2}$, the delayed displacement fulfills

$$x(t-\tau) \propto -v(t), \tag{33}$$

so that the feedback force

$$F_{fb} = -\frac{\Psi(x(t-\tau))}{m}x(t-\tau) \tag{34}$$

becomes proportional to $+v(t)$. This corresponds to negative damping, i.e., a force acting in phase with the velocity, which injects energy into the system and leads to amplification. This mechanism is consistent with classical descriptions of self-oscillatory systems [17].

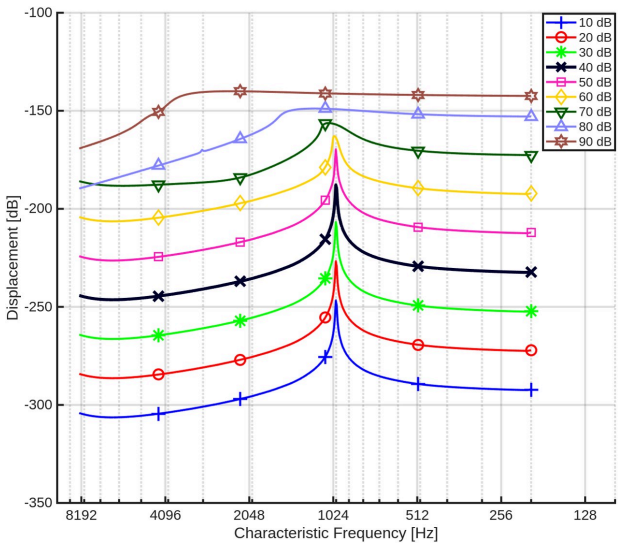

Figure 1: RMS displacement of 550 SLOs in the ensemble with nearest-neighbor coupling for a 1kHz sinusoidal stimulus at different SPLs, evaluated over 250 ms of steady state. Parameters: $\mu_i = 0$, $\beta_i = 1$, $k = 5$, $d = 5$, no noise, 600 oscillators logarithmically spaced between 200 Hz and 16 kHz.

This interpretation also agrees with the physiological description of cochlear mechanics by Lukashin et al. [18], where outer hair cells and the tectorial membrane provide active feedback to the basilar membrane. When the stimulus frequency matches the characteristic frequency, the system naturally operates close to the required phase condition, resulting in amplification. This suggests that an SLO can effectively describe a local cochlear partition near the bifurcation.

This behavior is illustrated in Figure 1, which shows the displacement response of the oscillator ensemble to a 1 kHz sinusoidal stimulus at different sound pressure levels. The model reproduces key cochlear features, including sharp frequency selectivity at low amplitudes, a shift of the peak response toward lower frequencies at higher amplitudes, and nonlinear compression. These characteristics are consistent with results from transmission line models [6] and experimental observations in mammalian cochleae [2].

Furthermore, Figure 2 demonstrates that an ensemble of coupled SLOs can reproduce a traveling-wave-like pattern. Importantly, this wave does not arise from a propagation of the stimulus through space. Instead, all oscillators are driven simultaneously, and the apparent traveling wave emerges from differences in phase and amplitude across the ensemble. Each oscillator responds with its own phase delay and gain, resulting in a spatial pattern resembling a traveling wave. Similar entrainment-induced wave patterns have been reported for coupled nonlinear oscillators such as the van der Pol system [19].

The transmission line model, due to its delay term, is inherently infinite-dimensional and can exhibit complex dynamical behavior, including chaos. In contrast, the reduced Stuart–Landau description is effectively two-dimensional near the Hopf bifurcation and captures the system's dominant deterministic dynamics.

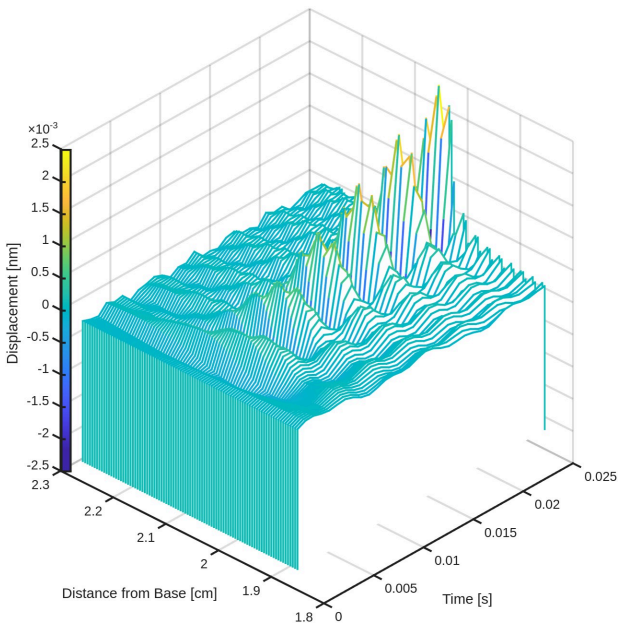

Figure 2: Displacement of 70 SLOs in the ensemble with nearest-neighbor coupling for a 1 kHz sinusoidal stimulus at 20 dB SPL before reaching steady state. Parameters: $\mu_i = 0$, $\beta_i = 1$, $k = 5$, $d = 5$, no noise.

However, a deterministic two-dimensional system cannot exhibit chaotic behavior. Chaos in the reduced model can only arise in the presence of stochastic perturbations [20]. As shown previously by Faber and Bozovic [20], the inclusion of intrinsic system noise in the reduced dynamics can lead to noise-induced chaos. This highlights that both descriptions—the infinite-dimensional transmission line model and its two-dimensional reduction—can exhibit complex dynamics, although the underlying mechanisms differ.

5. Conclusion

Overall, it has been shown that the SLO provides a reduced description of the transmission line model near the Hopf bifurcation. This reduction replaces the original delay differential equation by a significantly simpler ordinary or stochastic differential equation, while preserving key dynamical features such as frequency selectivity and nonlinear compression.

Furthermore, an ensemble of coupled SLOs offers an abstract yet insightful representation of cochlear partition dynamics. Interpreting the cochlea as a tonotopically organized array of coupled oscillators explains how spatially distributed phase and amplitude differences can give rise to traveling-wave-like behavior through entrainment, even when all oscillators are driven simultaneously.

At the same time, the nearest-neighbor coupling used in the model leads to distinct behavior at the boundaries of the oscillator ensemble. This raises the question of whether similar edge effects occur in the biological cochlea, and if not, which structural or dynamical mechanisms prevent them. Addressing this issue may be important for improving the biological realism of such models.

Finally, the reduction from an infinite-dimensional transmission line model to a two-dimensional system highlights an important conceptual point. While the full system can exhibit complex, potentially chaotic dynamics, the reduced deterministic SLO cannot do so on its own. However, as demonstrated, the inclusion of intrinsic noise can induce chaotic behavior in the reduced system. This raises the question of whether aspects of the original high-dimensional dynamics are effectively suppressed by the reduction and reintroduced through stochastic or external perturbations, or whether these represent fundamentally different mechanisms. Furthermore, certain observed phenomena may be explained only by chaotic dynamics, underscoring the potential relevance of chaos in cochlear modeling and motivating the use of SDE models.

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